Figure 4: Why oracle and OCS are close to fair static

Bottleneck / oracle explanation. Times are optical circuit/capacity reference model.

Workload	Method	Candidate	Hash	Bottleneck	Sender ms	Receiver ms	Resource ms	Same fair?
Qwen MMLU ML	fair universal static	random_regular_seed_11	bf8048120d	optical_resource	3.20	5.30	18.20	True
Qwen MMLU ML	prefill-informed OCS	random_regular_seed_11	bf8048120d	optical_resource	3.20	5.30	18.20	True
Qwen MMLU ML	oracle	random_regular_seed_11	bf8048120d	optical_resource	3.20	5.30	18.20	True
DeepSeek MMLU ML	fair universal static	random_regular_seed_11	bf8048120d	optical_resource	2.36	1.97	10.11	True
DeepSeek MMLU ML	prefill-informed OCS	random_regular_seed_27	1976e348a1	optical_resource	2.36	1.97	9.96	False
DeepSeek MMLU ML	oracle	random_regular_seed_27	1976e348a1	optical_resource	2.36	1.97	9.96	False
Qwen LiveCodeBench	fair universal static	random_regular_seed_11	bf8048120d	optical_resource	13.72	37.66	85.17	True
Qwen LiveCodeBench	prefill-informed OCS	random_regular_seed_2	22090398f3	optical_resource	37.66	13.72	84.32	False
Qwen LiveCodeBench	oracle	random_regular_seed_8	d2514c6e39	optical_resource	37.66	13.72	82.75	False